



DEPARTMENT OF MATHEMATICS - 29 November 2019

MATHEMATICAL MODELLING FOR INDUSTRY

MMI PRESENTATION

Instructor: Prof. S Sundar

By:
Karl Darryl Lewis
MA18C014

Contents

1	ASSIGNMENT 1: Image denoising	2
1.1	Image denoising techniques and MATLAB Simulation results	2
1.1.1	Linear diffusion	2
1.1.2	Non Linear Diffusion	2
1.1.3	Perona Malik Model	2
1.1.4	Catte-Lion Diffusion	3
1.2	Edge Enhancing Diffusion	3
1.3	Coherence Enhancing Diffusion	5
1.4	Comparison of different filters	5
1.5	Relationship between parameters σ, λ and T	6
2	ASSIGNMENT 2: Traffic flow modelling	7
2.1	Godunov scheme	7
3	ASSIGNMENT 3: Problem solving session	11
3.1	Caterer problem	11
3.2	Box office and cherry problem	14
4	ASSIGNMENT 4: Finite element method for Poisson equation	16
4.1	FEM formulation of Poisson's equation	16
4.2	Simulation results	19
5	ASSIGNMENT 5: Infectious disease model	22
5.1	The model	22
5.2	Simulation results	24

1 ASSIGNMENT 1: Image denoising

This assignment compares various types of diffusion filters. The techniques explored here are as follows: Linear diffusion filtering, Perona Malik model, Edge Enhancing Diffusion(EED) and Coherence Enhancing Diffusion.

The process of image enhancement broadly involves suppressing the noise while preserving the edges. Hence, basically it is a low pass filtering process. The physical phenomenon that imitates our requirement is that of diffusion. Cleaning of images uniformly over all directions is not very effective in image restoration. This is because the filter doesn't take into consideration the edges that need to be preserved.

1.1 Image denoising techniques and MATLAB Simulation results

1.1.1 Linear diffusion

Linear diffusion process can be classified into Linear Isotropic Diffusion and Linear Anisotropic diffusion.

Linear diffusion process in images is equivalent to the process of convolving the image with a 2D Gaussian kernel. The Gaussian kernel is the same as the one obtained by solving the heat equation. Hence the action of this kernel on the image is equivalent to the process of isotropic diffusion.

Anisotropic linear diffusion can be generalized from the isotropic linear diffusion by relaxing the requirement for invariance under rotation. This gives us the affine gaussian scale space. Now, this generalized filter lets us perform direction dependent cleaning.

1.1.2 Non Linear Diffusion

A PDE formulation for adaptive smoothing methods was first given by Perona and Malik. This is a nonlinear diffusion process which makes use of the local properties of the image. In their paper, they made use of scalar diffusivity and not a diffusivity tensor. Hence it is an isotropic process. This can be extended to an anisotropic diffusion process which makes use of a diffusion tensor instead of scalar diffusivity.

1.1.3 Perona Malik Model

This is a model used to avoid blurring and localization problems of linear diffusion filtering. This is an adaptive model. The diffusivity of the model is reduced at those locations where edges are more likely. The likelihood is measured using the quantity $|\nabla u|^2$. Visually, the performance of the PM model is very good. Edges are preserved. This is because edge detection and low pass filtering of noise happen in one single process. This model also has good contrast enhancement properties. However, this model is unstable and illposed.

1.1.4 Catte-Lion Diffusion

In Perona Malik diffusion scheme, sharp variation of noise is interpreted as edges. It is also unstable and illposed. To avoid this the diffusivity is replaced by a smoothened version. The smoothening of the diffusivity is carried out by convolving the diffusivity tensor with a Gaussian.



FIGURE: Simulation results of the Perona Malik scheme.

1.2 Edge Enhancing Diffusion

In the Perona Malik method, the process acts as a linear diffusion process in the interior of the edges, and hence removing noise. However, it fails to remove noise present at the edges. Therefore we go for Edge Enhancing Diffusion technique. In this technique smoothening of the image is carried out with simultaneous edge enhancement. There is contrast enhancement in anisotropic diffusion filters like in the case of Perona Malik model, provided the regularization parameter is not too large. Here, the two regions with almost constant grey value evolve which are separated by a steep edge. In this technique edge enhancement is caused by rapidly decreasing diffusivities. Intra-region smoothening is preferred over inter-region smoothening between two adjacent regions. The location of the edges remain unchanged over a long period of time. Hence determining the stopping time for cleaning is not an issue. After the process of contrast enhancement is complete, the steepness of edges decreases very slowly until gradient reaches a value where no backward diffusion is possible. After backward diffusion stops, the image quickly turns into a constant image.



FIGURE: MATLAB Simulation results of nonlinear anisotropic on "Lenna" image.

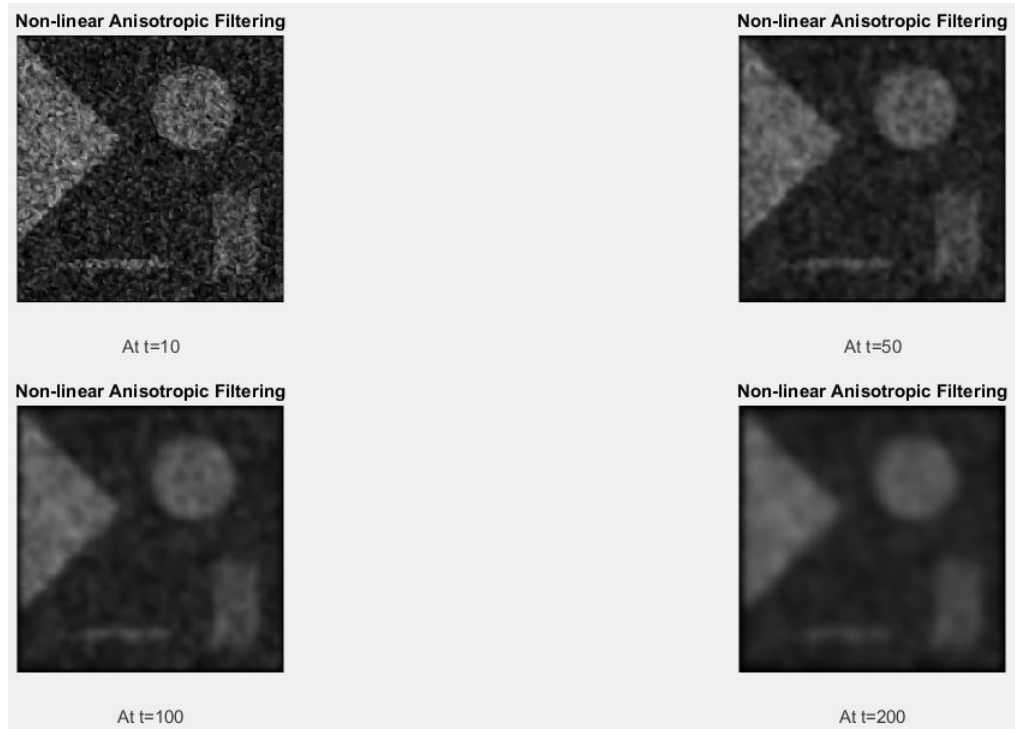


FIGURE: MATLAB Simulation results of nonlinear anisotropic on "synimgn2" image.

1.3 Coherence Enhancing Diffusion

In this technique, structure tensor information is used to design anisotropic scale spaces which enhance the coherence of flow like textures. This technique is very useful in applications like fingerprint analysis where the image contains flow like texture. It is also useful to have a multiscale simplification of the original image by embedding it into a scale-space in order to obtain a subsequently coarser, more global impression of the main flow-like structures. Such a scale-space should take into account the coherence of the structures by smoothing mainly along their preferred orientation instead of perpendicular to it.

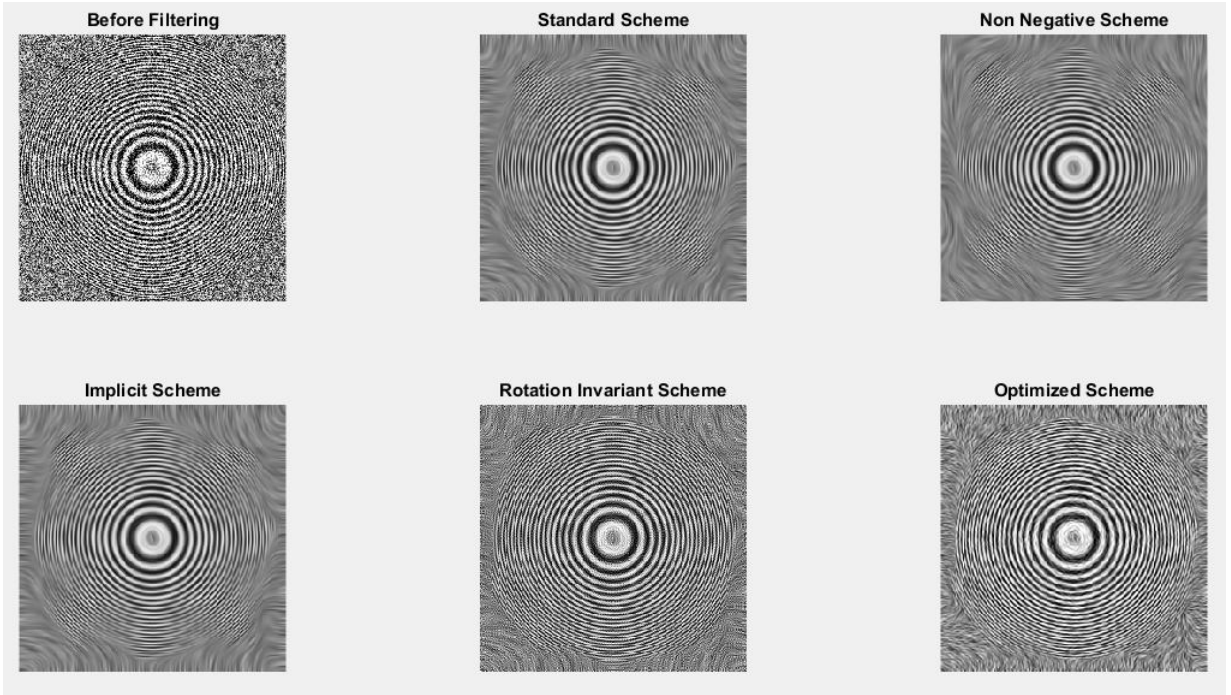


FIGURE: Simulation results of Coherence Enhanced Diffusion for different discretization schemes.

1.4 Comparison of different filters

In linear diffusion filtering, there is potential to remove all noise but the image becomes completely blurred. Edges are also smoothened along with the noise. Not only do the edges get smoothened but they also get dislocated. Once they are located at coarser scale, they have to be mapped back to their true location. This is hard to do.

In isotropic non linear diffusion like Perona Malik diffusion, the problem of dislocation of edges doesn't occur. However, in this process the diffusivity near the edges is very low. Thus noise near the edges is also preserved.

The PM diffusion is unstable and illposed. It recognizes sharp variations of noise as edges. The PMC method handles this problem by replacing the diffusivity by a smoothened version. The smoothening is carried out by convolving the scalar diffusivity with a Gaussian kernel.

Nonlinear anisotropic filters has advantages of both methods. We get good noise reduction performance

as well as stable edge structure. Since smoothening is allowed at the edges, the edges become rounded. The use of diffusion tensors instead of scalar-valued diffusivities puts us in a position to design real anisotropic diffusion processes which may reveal advantages at noisy edges. The fact that these filters are steered by the structure tensor instead of the regularized gradient allows to adapt them to more sophisticated tasks such as the enhancement of coherent flow-like structures. Thus CED is effective in applications like fingerprint scanning.

1.5 Relationship between parameters σ , λ and T

The parameter σ denotes the standard deviation of the Gaussian kernel. As σ increases the number of amount of averaging increases. Thus in linear filtering, the image becomes more blurred as σ increases. The parameter T denotes the stopping time for smoothening structures of order σ . The stopping time T is related to σ as follows: $T = \frac{\sigma^2}{2}$.

T is the optimal time to carry out the cleaning process. If the cleaning time is less than T , then the image is not cleaned properly. If the cleaning time is greater than T , then the image gets blurred. λ denotes the contrast parameter in Perona Malik diffusion scheme. It shows up in the expression for scalar diffusivity.

$$g(s^2) = \frac{1}{1 + \frac{s^2}{\lambda^2}}$$

As a result the expression for flux becomes,

$$\Phi(s) = \frac{s}{1 + \frac{s^2}{\lambda^2}}$$

When the gradient is less than λ , the flux is increasing in nature. Thus regions with gradient less than λ are diffused. Thus this region will have low contrast. Similarly, the region where gradient is greater than λ will have high contrast.

2 ASSIGNMENT 2: Traffic flow modelling

2.1 Godunov scheme

```
1  function ret = nf( u,v )
2
3  for i = 1:length(u)
4      if (fl(u(i)) >= 0 && fl(v(i))>=0)
5          ustar(i)=u(i);
6      else if (fl(u(i)) < 0 && fl(v(i))<0)
7          ustar(i)=v(i);
8      else if (fl(u(i)) >= 0 && fl(v(i))<0)
9          s=(f(v(i))-f(u(i)))/(v(i)-u(i));
10         if (s>=0)
11             ustar(i)=u(i);
12         else
13             ustar(i)=v(i);
14         end
15     else if ((fl(u(i)) < 0 && fl(v(i))<=0))
16         ret= 0;
17     end
18
19
20
21
22     end
23     end
24 end
25 end
26
27 ret =f(ustar);
28
29
```

FIGURE: Code for the Godunov scheme.

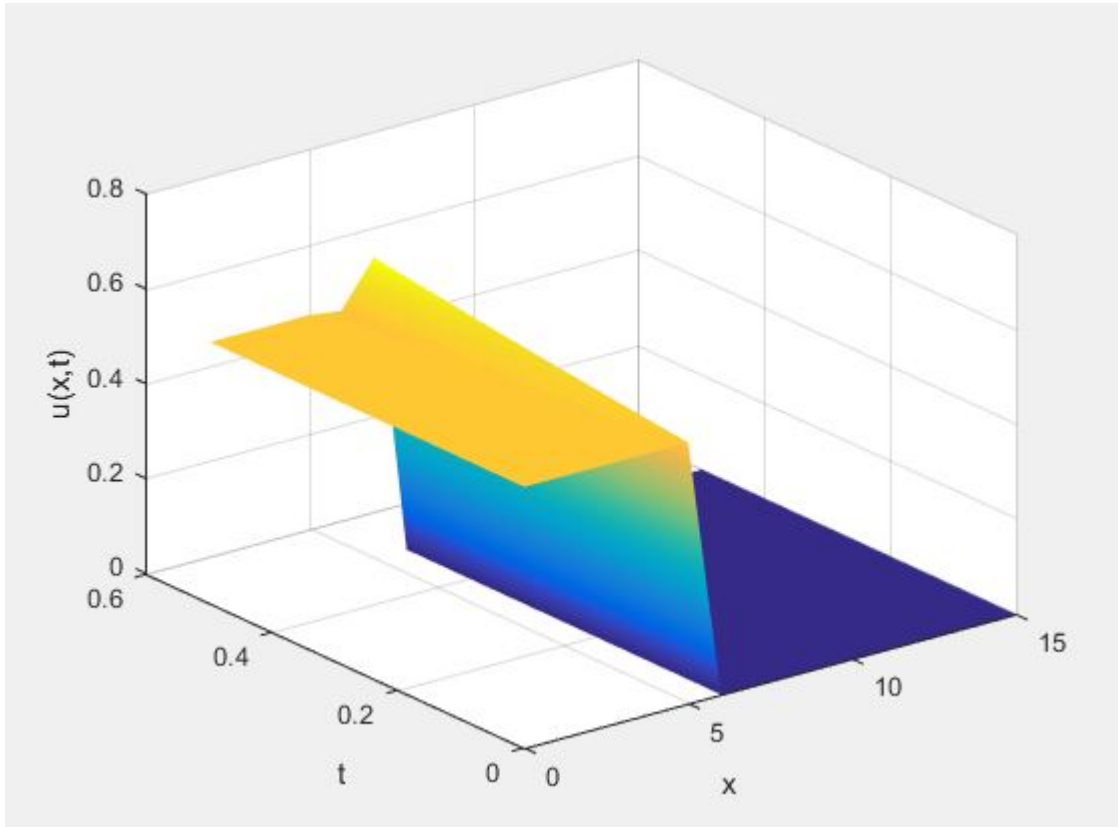


FIGURE: Simulation results of the Godunov scheme.

```

>> godunovtest
For t=1
  Columns 1 through 7
    0.5500    0.5500    0.5500    0.5500    0.5500    0.5651    0

  Columns 8 through 14
    0         0         0         0         0         0         0

  Columns 15 through 16
    0         0

For t=2
  Columns 1 through 7
    0.5500    0.5500    0.5500    0.5500    0.5492    0.5811    0

  Columns 8 through 14
    0         0         0         0         0         0         0

  Columns 15 through 16
    0         0

```

FIGURE: Simulation results of the Godunov scheme.

```

For t=3
  Columns 1 through 7
    0.5500    0.5500    0.5500    0.5500    0.5474    0.5980    0

  Columns 8 through 14
    0         0         0         0         0         0         0

  Columns 15 through 16
    0         0

For t=4
  Columns 1 through 7
    0.5500    0.5500    0.5500    0.5502    0.5445    0.6159    0

  Columns 8 through 14
    0         0         0         0         0         0         0

  Columns 15 through 16
    0         0

```

FIGURE: Simulation results of the Godunov scheme.

3 ASSIGNMENT 3: Problem solving session

3.1 Caterer problem

Caterer problem done in class:

$n=8$ $\{X_i\}_{i=1, \dots, 8}$

Fast laundry: 2 days (Napkins put on day 1 can be used on day 3)
 Slow laundry: 3 days (Napkins put on day 1 can be used on day 4)

Fast laundry cost: 25 Rs (b) Slow laundry cost: 15 Rs (c)

$\{Y_i\}_{i=1, \dots, 6} \leftarrow$ No. of napkins given to fast laundry at the end of day i ;
 $\{Z_i\}_{i=1, \dots, 5} \leftarrow$ No. of napkins given to slow laundry at the end of day i .

Let Z_i 's be 200, 80, 140, 90, 100, 150, 80, and 180.
 Let cost of new napkins be Rs 100 (=a)

Minimize: $100(X_1 + X_2 + \dots + X_8) + 25(Y_1 + Y_2 + \dots + Y_6) + 15(Z_1 + Z_2 + \dots + Z_5)$

Day 1 demand: $X_1 \geq 200$

Day 2 demand: $200(X_1 - 200) + X_2 \geq 80 \Rightarrow X_1 + X_2 \geq 280$

Day 3 demand: $(X_1 + X_2 - 280) + X_3 + Y_1 \geq 140 \Rightarrow X_1 + X_2 + X_3 + Y_1 \geq 420$

Day 4 demand: $(X_1 + X_2 + X_3 + Y_1 - 420) + X_4 + Y_2 + Z_1 \geq 90$
 $\Rightarrow X_1 + X_2 + X_3 + X_4 + Y_1 + Y_2 + Z_1 \geq 510$

Day 5 demand: $(X_1 + X_2 + X_3 + X_4 + Y_1 + Y_2 + Z_1 - 510) + Y_3 + Z_2 \geq 100$
 $\Rightarrow X_1 + X_2 + X_3 + X_4 + X_5 + Y_1 + Y_2 + Y_3 + Z_1 + Z_2 \geq 610$

Day 6 demand: $X_1 + X_2 + X_3 + X_4 + X_5 + X_6 + Y_1 + Y_2 + Y_3 + Y_4 + Z_1 + Z_2 + Z_3 \geq 610 + 150 = 760$

Day 7 demand: $X_1 + X_2 + X_3 + X_4 + X_5 + X_6 + X_7 + Y_1 + Y_2 + Y_3 + Y_4 + Y_5 + Z_1 + Z_2 + Z_3 + Z_4 \geq 840$

Day 8 demand: $X_1 + X_2 + X_3 + X_4 + X_5 + X_6 + X_7 + X_8 + Y_1 + Y_2 + Y_3 + Y_4 + Y_5 + Y_6 + Z_1 + Z_2 + Z_3 + Z_4 + Z_5 \geq 1020$

There are constraints in picking used napkins to laundry:
 $Y_1 + Z_1 \leq 200, Y_2 + Z_2 \leq 80, Y_3 + Z_3 \leq 140, Y_4 + Z_4 \leq 90, Y_5 + Z_5 \leq 100, Y_6 \leq 150$

~~Lower bounds:~~ $X_i, Y_i, Z_i \geq 0$

* We use Imp. (3) to solve the problem.

Obtained results: $X_1 = 238, X_2 = 461, X_3 = X_4 = X_5 = 2X_6 = \dots = X_8 = 20$
 $Y_1 = 140, Y_2 = 30, Y_3 = 20, Y_4 = 60, Y_5 = 50, Y_6 = 130$
 $Z_1 = 60, Z_2 = 50, Z_3 = 90, Z_4 = 30, Z_5 = 50$

19 variables
 14 constraints
 19 lower bounds.

FIGURE: LPP formulation of the caterers problem.

```

1      % Constraint matrix
2 -    A=[-1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0;
3          -1 -1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0;
4          -1 -1 -1 0 0 0 0 0 -1 0 0 0 0 0 0 0 0 0 0 0;
5          -1 -1 -1 -1 0 0 0 0 -1 -1 0 0 0 0 -1 0 0 0 0 0;
6          -1 -1 -1 -1 -1 0 0 0 -1 -1 -1 0 0 0 -1 -1 0 0 0 0;
7          -1 -1 -1 -1 -1 -1 0 0 -1 -1 -1 -1 0 0 -1 -1 -1 0 0 0;
8          -1 -1 -1 -1 -1 -1 -1 0 -1 -1 -1 -1 -1 0 -1 -1 -1 -1 0 0;
9          -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1;
10         0 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 1 0 0 0 0;
11         0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 1 0 0 0;
12         0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 1 0 0;
13         0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 1 0;
14         0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 1;
15         0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0];
16
17 -    % RHS of constraints
18     b=[-200 -280 -420 -510 -610 -760 -840 -1020 200 80 140 90 100 150];
19
20 -    % Equality constraints (There are none)
21     Aeq=[];
22     beq=[];
23
24 -    % Lower bounds
25     lb=[0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0];
26
27 -    % Objective function
28     f=[100 100 100 100 100 100 100 100 100 100 25 25 25 25 25 25 15 15 15 15];
29
30 -    % Optimization results
31     [x,minimumfval]=linprog(f,A,b,Aeq,beq,lb)

```

FIGURE: Caterer problem code

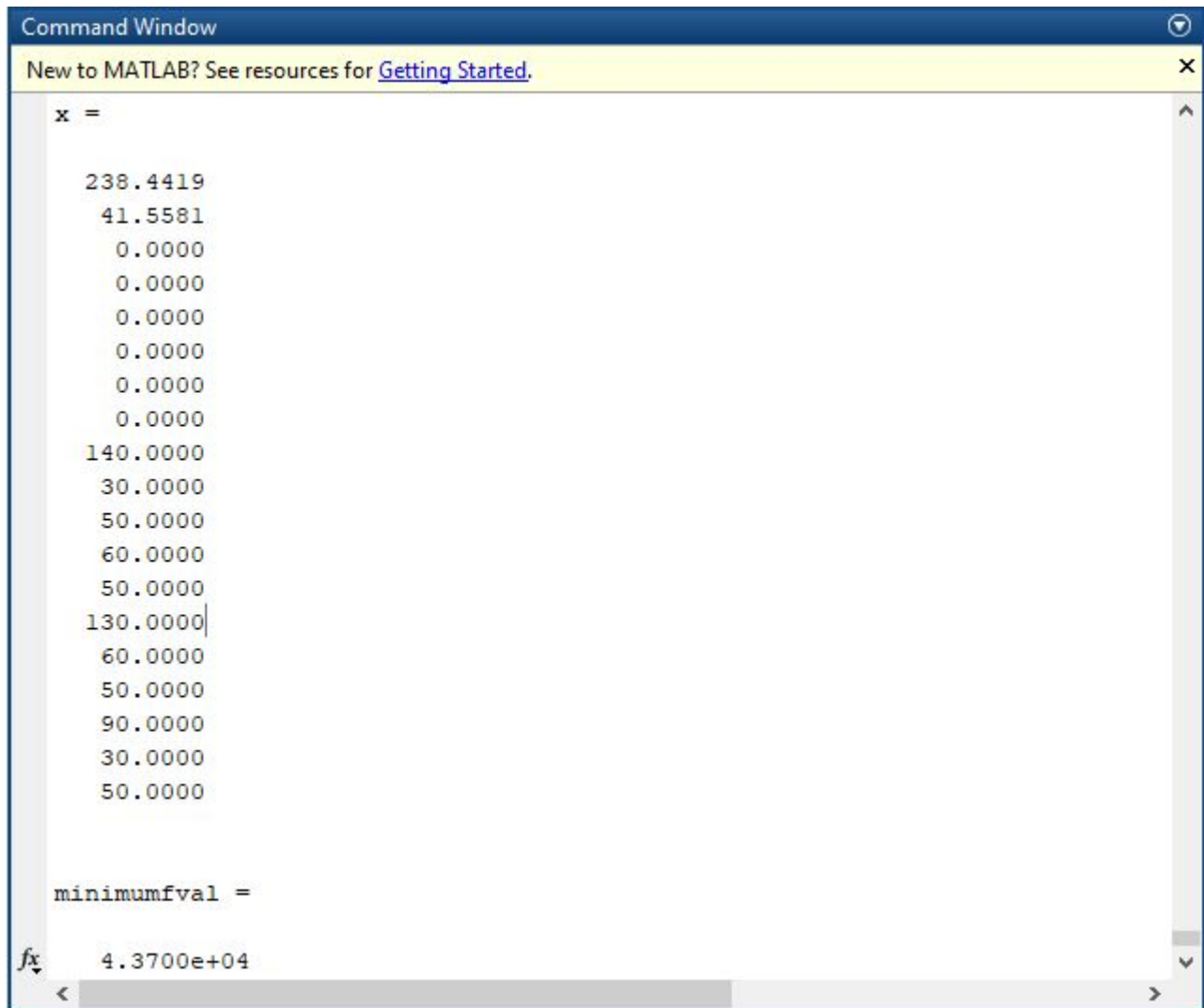


FIGURE: Optimization results for caterer problem.

3.2 Box office and cherry problem

Number of cherries in a can: $r \leftarrow$ cherry, can $\rightarrow r, h$

Upper bound $\approx 0.74 \times \frac{\text{Volume of can}}{\text{Volume of cherry}}$ Kepler's conjecture was proved in 1998

Lower bound $\approx$: Find how many circles of radius r can be packed in a circle of radius R . Divide the height of the can by $2r$. Product of these two quantities gives a lower bound.

Generalizations: Spheres in cuboid, cuboids in cuboid, polyhedra in spheres, circle in circle (proved for $n \leq 13$ & $n \geq 19$), circle in square.

* Lower bounds are usually easy to compute.

Box office problem

$2n$ people lined up at a ticket office. n have \$5, n have \$10 bills.

Box office has no cash when it opens. Ticket cost: \$5.

What is the probability that no customer waits for change?

$$\frac{\binom{2n}{n} - \binom{2n}{n+1}}{\binom{2n}{n}} \approx \frac{1}{n+1} \quad C_n \leftarrow h^n \text{ Catalan number}$$

Generalization:

$3n$ people n \$5Rs n \$10Rs n have Rs 15 note.

Can't do it because we have to give back $2n$ \$5Rs to people who give \$10Rs notes & n change to people who give 15Rs. But we have only n \$5Rs notes.

$5n$ people; 2 $3n$ people have Rs 5, n have Rs 10, n have Rs 15.

Still not possible! Ends at $(5n, 0)$ (trajectory)

Give the trajectory as Rs 5: -1 Rs 10: $+1$ Rs 15: $+2$.

FIGURE: Cherry problem and box office problem.

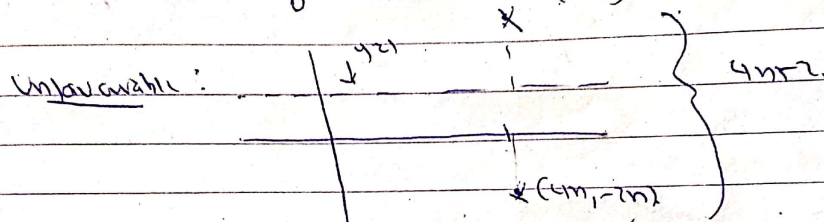
One possible generalization of box office problem

Consider $4n$ people to be in a queue

n have 10\$ bills, $3n$ have 5\$ bills

Transactions: +1 for 10\$ -1 for 5\$

Total number of trajectories: $\binom{4n}{n}$



$\therefore 2n+1$ max ascents & $2n+1$ descents

$$\text{No. of descents} = 3n - (2n+1) = n-1$$

$$\text{No. of ascents} = n + (2n+1) = 3n+1$$

$$\therefore \text{No. of unfavorable outcomes} = \binom{4n}{n-1}$$

$$\therefore \text{Probability of no queue} = \frac{\binom{4n}{n} - \binom{4n}{n-1}}{\binom{4n}{n}}$$

(However no Catalan numbers show up!)

FIGURE: Box office problem.

4 ASSIGNMENT 4: Finite element method for Poisson equation

4.1 FEM formulation of Poisson's equation

Karl D Lewis
MA18C014

Task 4

Poisson's equation

$$\frac{\partial^2 u}{\partial x^2} + f = 0 \quad u(a) = u(b) = 0 \leftarrow \text{Dirichlet boundary condition}$$

Weak formulation

To find the weak formulation, we multiply by an appropriate test function v and integrate over the domain

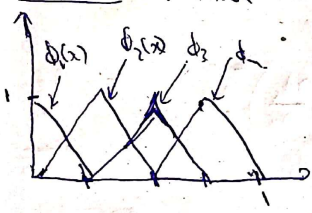
$$\frac{\partial^2 u}{\partial x^2} + f = 0 \Rightarrow \int_a^b v \left(\frac{\partial^2 u}{\partial x^2} + f \right) dx = 0 \quad \forall v \in \text{Test functions}$$

$$\Rightarrow \left[- \int_a^b \frac{\partial v}{\partial x} \frac{\partial u}{\partial x} dx + \int_a^b v f dx = 0 \right] \quad \text{Weak formulation.}$$

Let $\{ \phi_i(x) : i=1, \dots, n \}$ s.t. $u(x) \approx \sum u_i \phi_i(x)$

where $\{ \phi_i(x) : i=1, \dots, n \}$ form a basis.

Basis functions : choose $a=0, b=1$



$$\Rightarrow - \sum u_i \int_a^b \frac{\partial \phi_i}{\partial x} \frac{\partial \phi_j}{\partial x} dx + \int_a^b \phi_j f dx = 0 \quad \forall \phi_j \in \text{Test functions } j=1, \dots, n$$

$$\therefore - \sum u_i \left(\int_a^b \frac{\partial \phi_i}{\partial x} \frac{\partial \phi_j}{\partial x} dx \right) + \int_a^b \phi_j f dx = 0 \quad \forall \phi_j \in \text{Basis } j$$

FEM FORMULATION

$$A_{ij} = \frac{1}{x_i - x_{i-1}} + \frac{1}{x_{i+1} - x_i} \quad b_j = \left(\frac{x_{i+1} - x_{i-1}}{2} \right) f(x_i)$$

We solve $Au=b$ in MATLAB. The solution gives us the coefficients u_i in $u \approx \sum u_i \phi_i$

$\therefore u(x)$ can be reconstructed

CS Scanned with CamScanner

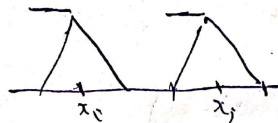
FIGURE: FEM formulation

Solving Poisson's equation using the piecewise linear basis

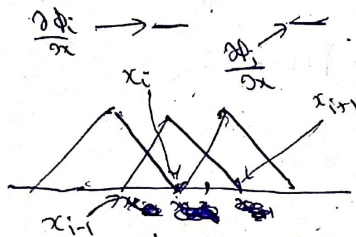
We have $A_{ij} = \int_0^1 \frac{\partial \phi_j}{\partial x} \frac{\partial \phi_i}{\partial x} dx$ for $j=1, 2, \dots, n-1$
 (∵ $u(0)=u(1)=0$, $j=0$ & $j=n$ don't matter)

$$b_j = \int_0^1 \phi_j f dx$$

$\{\phi_j\}$ is sparse because if i, j are not neighbours, then $A_{ij} = 0$



← No overlap $\Rightarrow \frac{\partial \phi_i}{\partial x} \frac{\partial \phi_j}{\partial x} = 0$
 $\Rightarrow A_{ij} = 0$



$$\therefore A_{ij} = \begin{cases} 0 & |i-j| > 1 \\ \text{non-zero} & |i-j| \leq 1 \end{cases}$$

$$A_{ii} = \int_0^1 \left(\frac{\partial \phi_i}{\partial x} \right)^2 dx = \int_{x_{i-1}}^{x_i} \left(\frac{\partial \phi_i}{\partial x} \right)^2 dx + \int_{x_i}^{x_{i+1}} \left(\frac{\partial \phi_i}{\partial x} \right)^2 dx$$

$$= \left(\frac{1}{x_i - x_{i-1}} \right)^2 \int_{x_{i-1}}^{x_i} dx + \left(\frac{1}{x_{i+1} - x_i} \right)^2 \int_{x_i}^{x_{i+1}} dx$$

$$= \left(\frac{1}{x_i - x_{i-1}} + \frac{1}{x_{i+1} - x_i} \right)$$

$$A_{ii} A_{i(i+1)} = \int_0^1 \left(\frac{\partial \phi_i}{\partial x} \right) \left(\frac{\partial \phi_{i+1}}{\partial x} \right) dx$$

$$= \int_{x_i}^{x_{i+1}} \left(\frac{\partial \phi_i}{\partial x} \right) \left(\frac{\partial \phi_{i+1}}{\partial x} \right) dx$$

$$= \frac{-1}{x_{i+1} - x_i} \cdot \frac{1}{x_{i+1} - x_i}$$

$$= - \int_{x_i}^{x_{i+1}} \frac{1}{(x_{i+1} - x_i)^2} dx = \frac{-1}{x_{i+1} - x_i}$$



Scanned with
CamScanner

FIGURE: FEM formulation

Actual solution of $U'' + 2 = 0$ $U(0) = 0 = U(1)$

$$\frac{d^2 U}{dx^2} + 2 = 0$$

Here $\frac{d}{dx} = \frac{d}{dx}$

$$\frac{d^2 U}{dx^2} + 2 = 0 \Rightarrow \frac{d^2 U}{dx^2} = -2$$

$$\Rightarrow \frac{dU}{dx} = -2x + C_1$$

$$\Rightarrow U = -x^2 + C_1 x + C_2$$

$$U(0) = 0 \Rightarrow 0 = C_2$$

$$U(1) = 0 \Rightarrow 0 = -1 + C_1 \Rightarrow C_1 = 1$$

$\therefore$ The actual solution is $U(x) = -x^2 + x$

From the matlab code we observe that the numerical solution matches with the actual solution at all grid points.

FIGURE: FEM formulation

4.2 Simulation results

```
1 - n=100
2 - x=rand(n-1,1)
3 - x=[0;sort(x);1]
4 - A=zeros(n-1)
5 - A=A+diag(1./(x(3:end)-x(2:end-1))+1./(x(2:end-1)-x(1:end-2)))
6 - A=A+diag(1./(x(3:end-1)-x(2:end-2)),1)
7 - A=A+diag(1./(x(3:end-1)-x(2:end-2)),-1)
8 - A=A-2*diag(1./(x(3:end-1)-x(2:end-2)),-1)
9 - A=A-2*diag(1./(x(3:end-1)-x(2:end-2)),1)
10 - b=2*0.5*(x(3:end)-x(1:end-2))
11 - u=A\b
12 - |
13 - % Numerical solution
14 - y(:,1)=[0;u;0]
15 - % Actual solution
16 - y(:,2)=-x.*x+x
17 -
18 - plot(x,y(:,1),'-o')
19 - hold on
20 - plot(x,y(:,2),'-x')
21 - title('Comparison of actual and numerical solution')
```

FIGURE: Code for FEM modelling of Poisson equation.

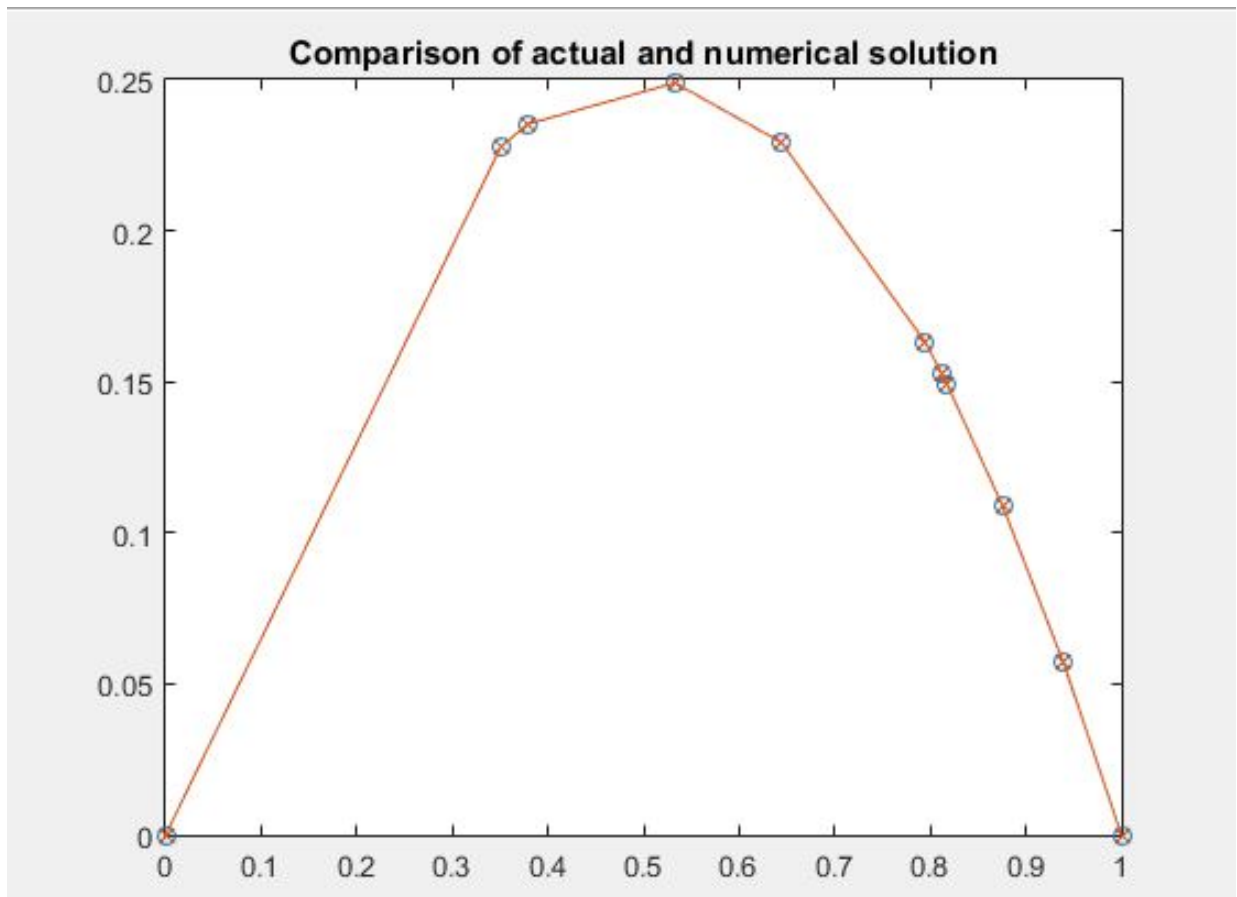


FIGURE: Numerical solution computed via FEM vs true solution for 10 random points.

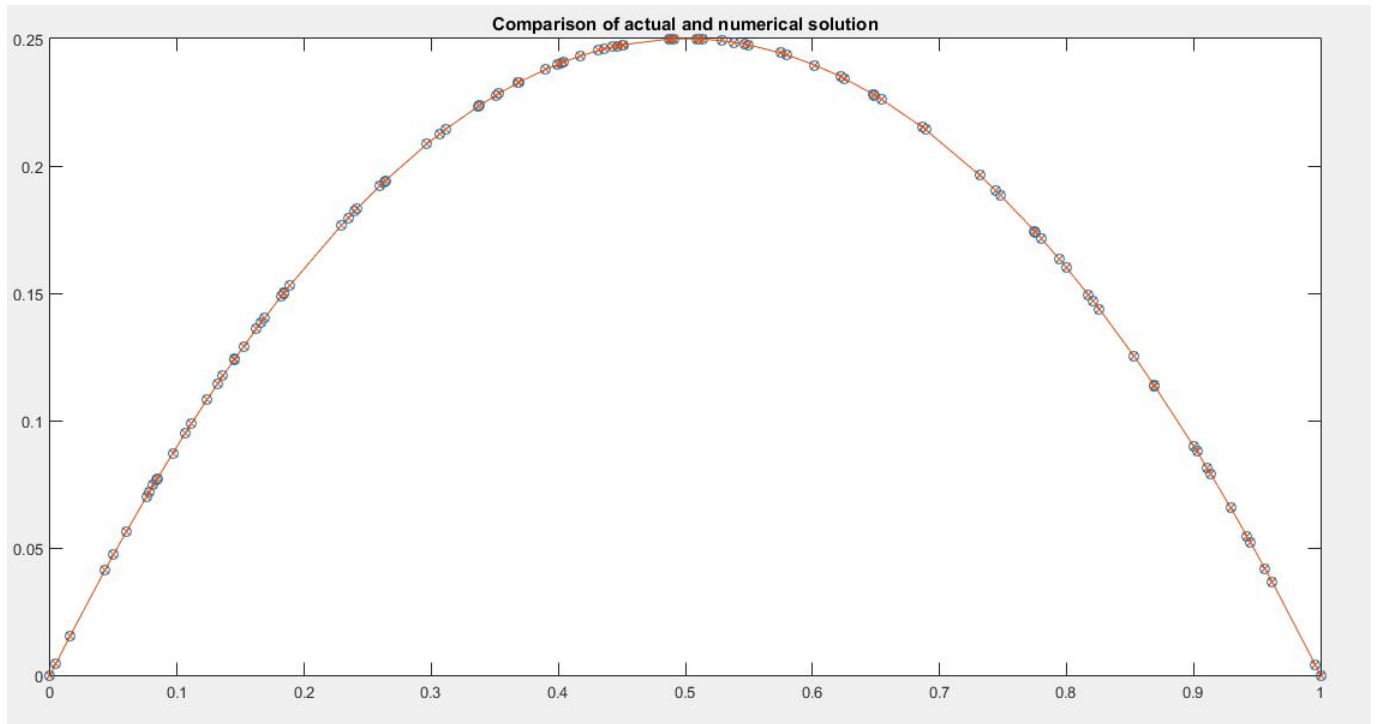


FIGURE: Numerical solution computed via FEM vs true solution for 100 random points.

5 ASSIGNMENT 5: Infectious disease model

5.1 The model

Mathematical modelling of infectious disease

Factors:

- 1) Concentration of pathogen multiplying antigens $V(t)$
- 2) Concentration of antibodies $F(t)$
- 3) Concentration of Plasma cells $C(t)$ (carries & produces antibodies)
- 4) Relative characteristic of affected organ $m(t)$
 $\leftarrow$ copy virus multiplication

$\frac{dV}{dt} = \lambda V - \rho F V$
 $\leftarrow$ This term is for the antigens neutralized by the antibodies
 $\leftarrow$ connected with the probability for an antigen to be neutralized after encountering an antibody.

$\Rightarrow \frac{dV}{dt} = \lambda V - \rho F V$ $\leftarrow$ This equation describes change in number of antigens in an organism.

Growth of plasma cells:

$\frac{dC}{dt} = qKF(t-\gamma)V(t-\gamma) - M(C-C^*)$
 $\gamma \rightarrow$ time of junction of the cascade of plasma cells.
 $qK \leftarrow$ probability of antigen-antibody collision
 $M \leftarrow$ coefficient equal to inverse of plasma cell life time.

$\Rightarrow \frac{dC}{dt} = -M(C-C^*) + q(m)KV(t-\gamma)F(t-\gamma)$ $\leftarrow$ delay differential equation, - (2)

Balance of the number of antibodies reacting with antigens:

$\frac{dF}{dt} = \beta C - \gamma F V - aF$
 $\beta C dt \leftarrow$ Generation of antibodies by plasma cells during time dt .
 $\beta \leftarrow$ rate of antibody production by one plasma cell.
 $\gamma F V dt \leftarrow$ decrease in the number of antibodies during the time dt due to binding with the antigens.
 $\rho F V dt \leftarrow$ No of antigens neutralized by antibodies.
 $\gamma \leftarrow$ one antigen requires γ antibodies.
 $aF dt \leftarrow$ Drop in antibody population due to aging.
 $a \leftarrow$ coefficient inversely proportional to decay of antibodies.

$\Rightarrow \frac{dF}{dt} = \beta C - \gamma F V - aF$ - (1)

FIGURE: Infectious disease model.

The equations described till now do not account for weakening of an organism during disease due to decrease in the activity of organs providing the delivery of immunologic materials:-

Hypothesis: Productivity of such an organ depends on the degree of damage of the target organ.

$$\frac{dm}{dt} = \sigma V - \eta m$$

$\sigma V \leftarrow$ degree of damage to an organ due to antigen.

$\eta m \leftarrow$ the β caused by recovery activity of the organism.

Take $m^* = 0.1$ $C^* = 10$ $V(0) = V_0$ $F(0) = F_0 = 10$ $q(0) = 1$.

① Subclinical form

$$d=8 \quad p=10 \quad \beta=0.17 \quad \gamma p=8 \quad \mu=0.5 \quad K=10^4 \quad \sigma=10$$

$$\eta=0.12 \quad \tau=0.5$$

$$V_0=10^{-5}, 10^{-3}.$$

② Acute form

$$d=2 \quad p=0.8 \quad \beta=0.17 \quad \gamma p=8 \quad \mu=0.5 \quad K=10^5 \quad \sigma=10$$

$$\eta=0.12 \quad \tau=0.5$$

$$V_0=10^{-5}, 10^{-7}, 10^{-9}.$$

③ Chronic form

$$d=1 \quad p=0.8 \quad \beta=0.17 \quad \gamma p=8 \quad \mu=0.5 \quad K=10^3 \quad \sigma=10$$

$$\eta=0.17 \quad \tau=0.5$$

④ Lethal outcome

$$d=1.54 \quad p=0.77 \quad \beta=0.15 \quad \gamma p=8 \quad \mu=0.5 \quad K=880 \quad \sigma=12$$

$$\eta=0.12 \quad \tau=0.5$$

FIGURE: Infectious disease model.

5.2 Simulation results

```

43 % 4th order RK Method implementation
44 for i=1:(length(t)-1)
45     k1v = fV(t(i),V(i),C(i),F(i),m(i));
46     k1f = fF(t(i),V(i),C(i),F(i),m(i));
47     k1c = fC(t(i),V(i),C(i),F(i),m(i));
48     k1m = fm(t(i),V(i),C(i),F(i),m(i));
49
50
51     k2v = fV(t(i)+0.5*h,V(i)+0.5*h*k1v,C(i)+0.5*h*k1c,F(i)+0.5*h*k1f,m(i)+0.5*h*k1m);
52     k2f = fF(t(i)+0.5*h,V(i)+0.5*h*k1v,C(i)+0.5*h*k1c,F(i)+0.5*h*k1f,m(i)+0.5*h*k1m);
53     k2c = fC(t(i)+0.5*h,V(i)+0.5*h*k1v,C(i)+0.5*h*k1c,F(i)+0.5*h*k1f,m(i)+0.5*h*k1m);
54     k2m = fm(t(i)+0.5*h,V(i)+0.5*h*k1v,C(i)+0.5*h*k1c,F(i)+0.5*h*k1f,m(i)+0.5*h*k1m);
55
56     k3v = fV(t(i)+0.5*h,V(i)+0.5*h*k2v,C(i)+0.5*h*k2c,F(i)+0.5*h*k2f,m(i)+0.5*h*k2m);
57     k3f = fF(t(i)+0.5*h,V(i)+0.5*h*k2v,C(i)+0.5*h*k2c,F(i)+0.5*h*k2f,m(i)+0.5*h*k2m);
58     k3c = fC(t(i)+0.5*h,V(i)+0.5*h*k2v,C(i)+0.5*h*k2c,F(i)+0.5*h*k2f,m(i)+0.5*h*k2m);
59     k3m = fm(t(i)+0.5*h,V(i)+0.5*h*k2v,C(i)+0.5*h*k2c,F(i)+0.5*h*k2f,m(i)+0.5*h*k2m);
60
61     k4v = fV(t(i)+h,V(i)+h*k3v,C(i)+h*k3c,F(i)+h*k3f,m(i)+h*k3m);
62     k4f = fF(t(i)+h,V(i)+h*k3v,C(i)+h*k3c,F(i)+h*k3f,m(i)+h*k3m);
63     k4c = fV(t(i)+h,V(i)+h*k3v,C(i)+h*k3c,F(i)+h*k3f,m(i)+h*k3m);
64     k4m = fV(t(i)+h,V(i)+h*k3v,C(i)+h*k3c,F(i)+h*k3f,m(i)+h*k3m);
65
66 %Updating step
67 V(i+1) = V(i) + (1/6) * (k1v+2*k2v+2*k3v+k4v) *h;
68 F(i+1) = F(i) + (1/6) * (k1f+2*k2f+2*k3f+k4f) *h;
69 C(i+1) = C(i) + (1/6) * (k1c+2*k2c+2*k3c+k4c) *h;
70 m(i+1) = m(i) + (1/6) * (k1m+2*k2m+2*k3m+k4m) *h;
71 end
72 plot(t,V)

```

FIGURE: Code for the fourth order Runge Kutta method.

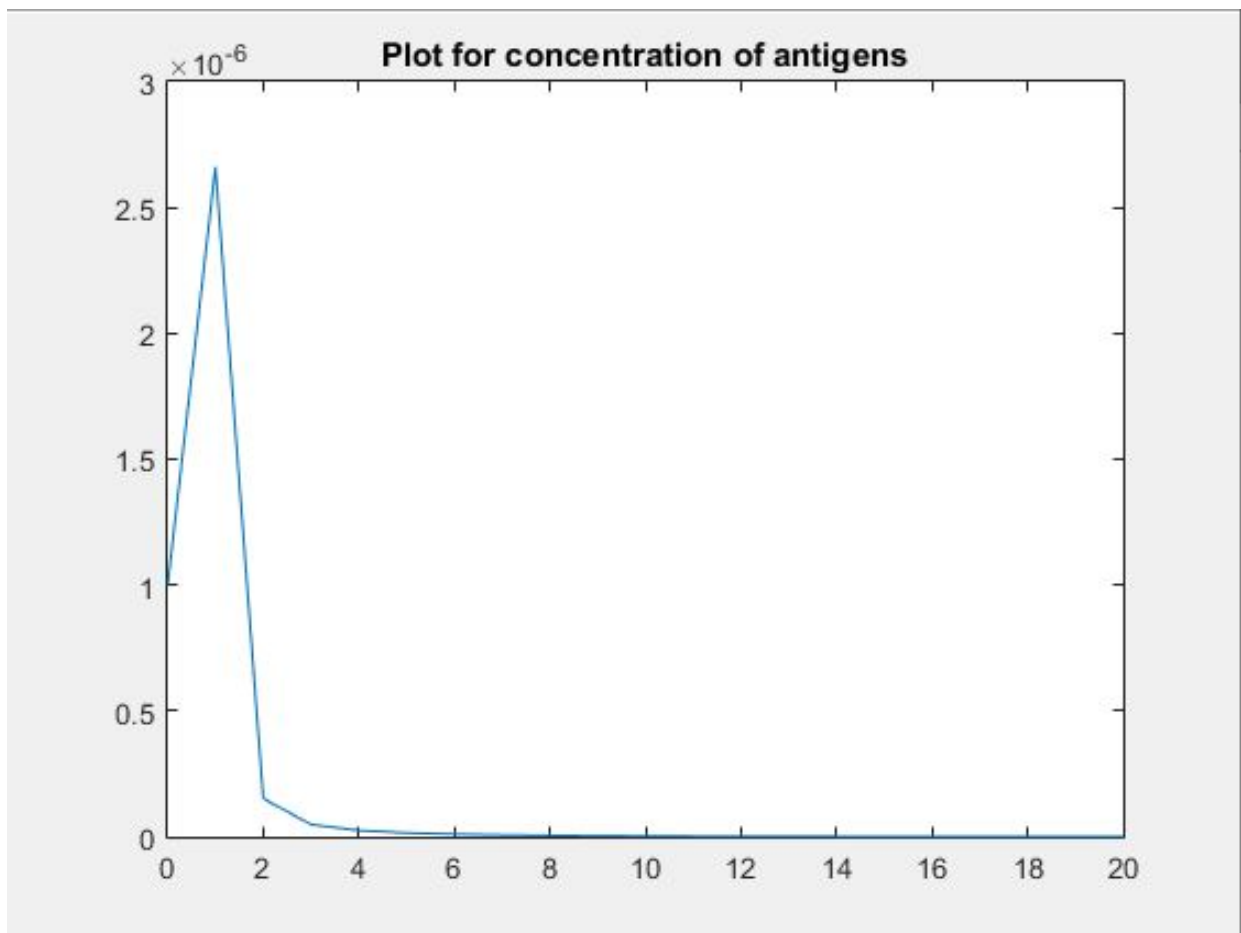


FIGURE: Infectious disease model.

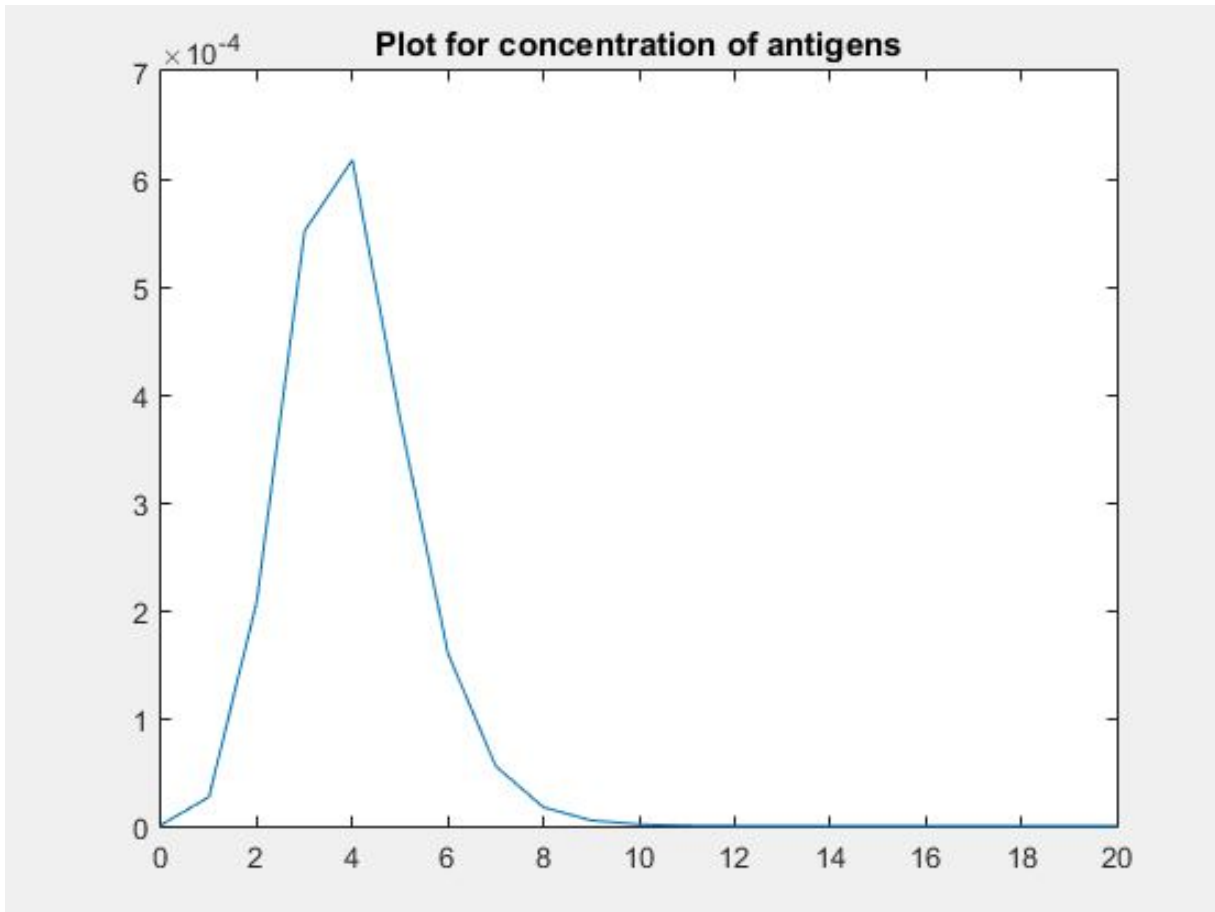


FIGURE: Infectious disease model.